

6 (**long**) A circular jet of water impinges onto an axi-symmetric deflector as shown in Fig. 3. The velocity of the circular jet is V_1 and its cross-sectional area is A_1 . After deflection through 180° to form an annular jet of cross-sectional area A_2 , the velocity is V_2 .

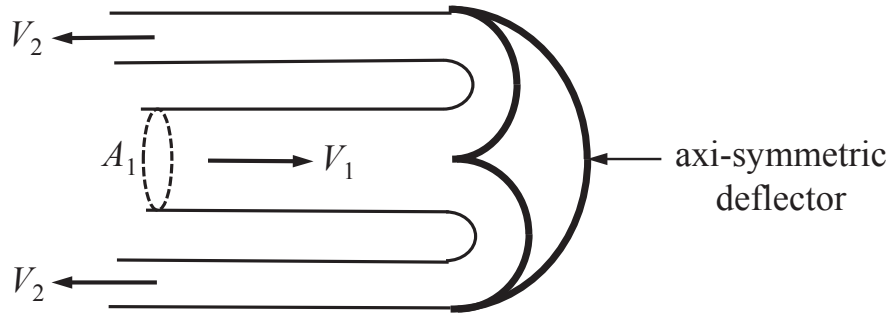


Fig. 3

(a) With the deflector stationary, and assuming the flow to be inviscid, give the relationship between the velocities V_1 and V_2 stating clearly the physical basis for your reasoning. [4]

(b) Calculate the relationship between the areas A_1 and A_2 stating clearly your assumptions. [5]

(c) Show that the force F on the deflector is given by

$$F = 2\rho QV_1$$

where ρ and Q are the density and the volume flow rate of the water, respectively. [7]

(d) If now the deflector is moving to the right with velocity U (assuming $U < V_1$), what is the force exerted by the water on the deflector? [7]

(e) Show that the maximum power is extracted from the water jet when the deflector velocity U equals $V_1/3$. State clearly your assumptions. [7]